



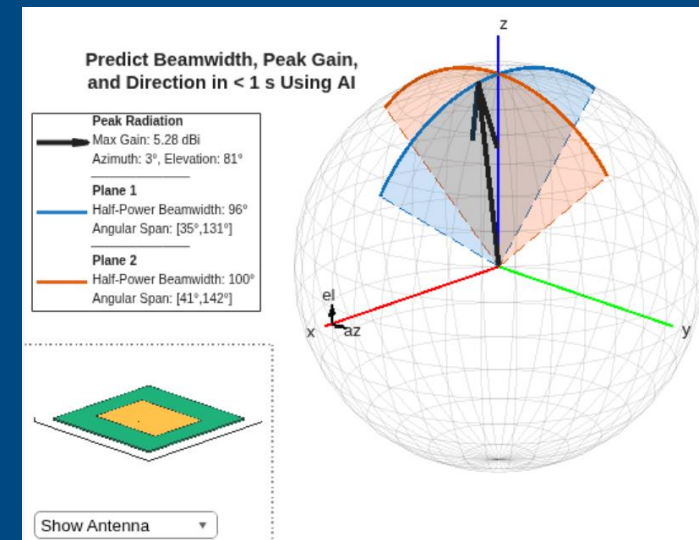
AP-S/URSI — Technical Workshop HD-7

AI in Antenna Design and Analysis: From Pretrained Models to System-Level Integration

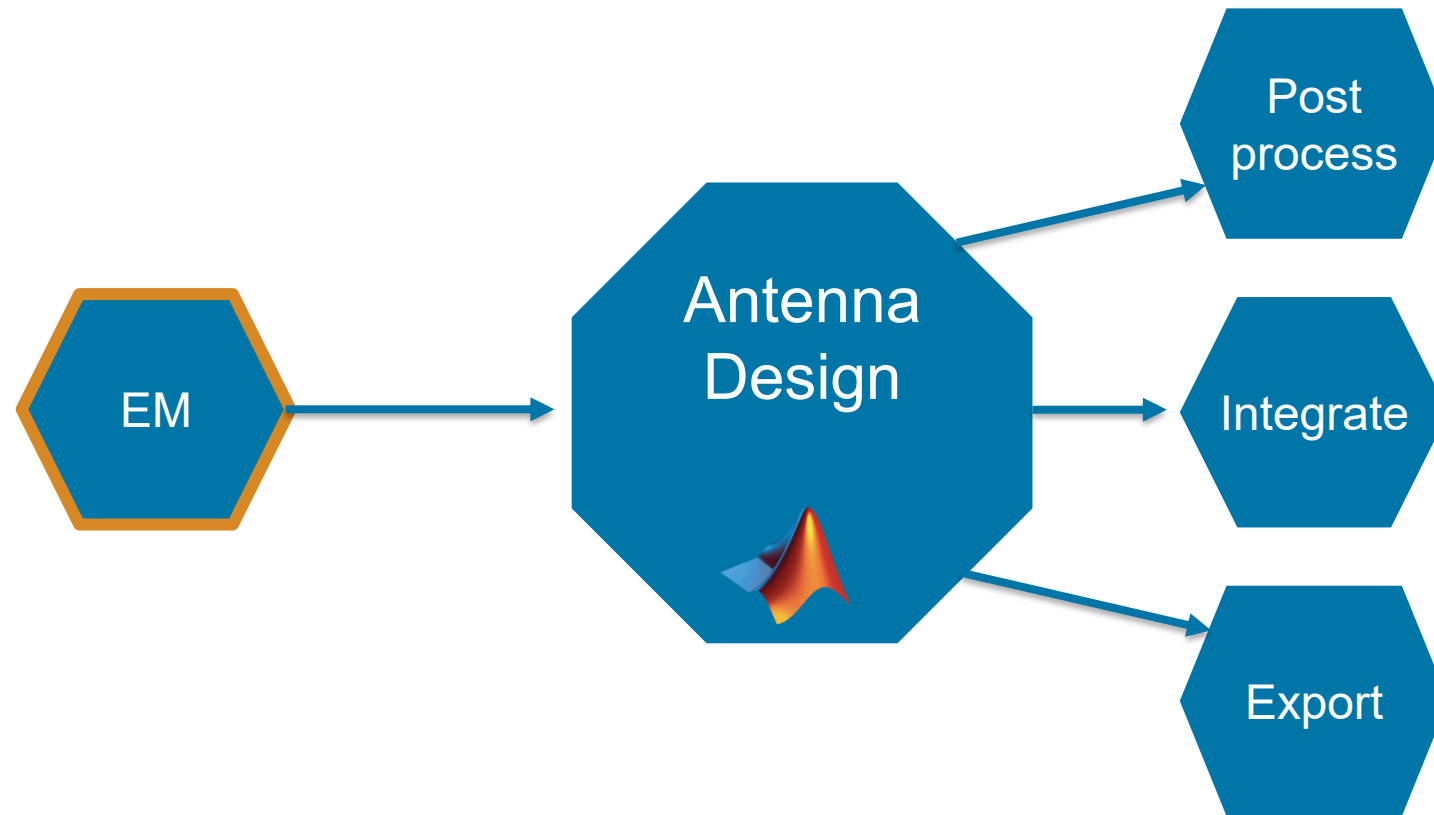
Part 1 — Pretrained AI Antennas for Design Space Exploration and AutoML

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MathWorks | RF/EM Group



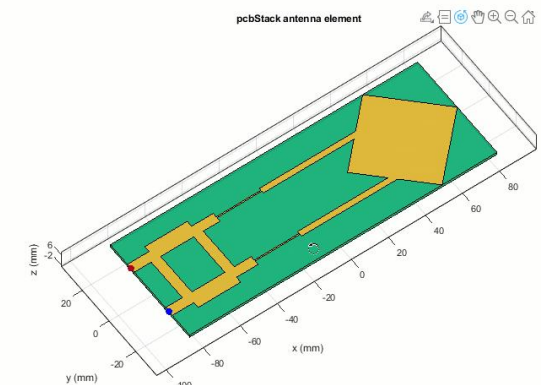
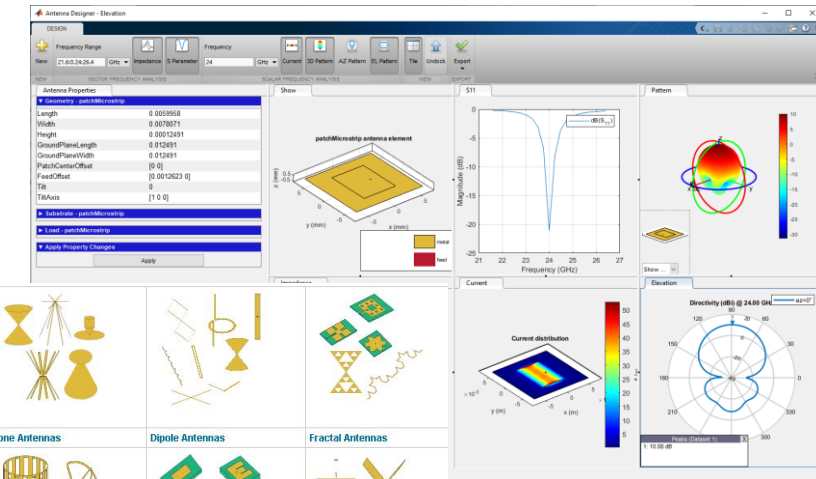
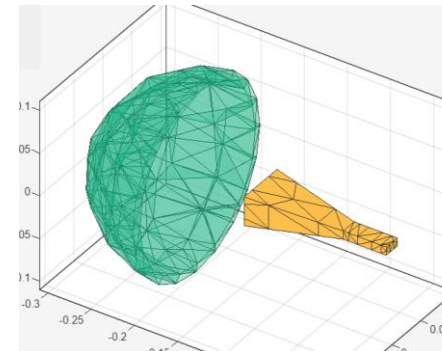
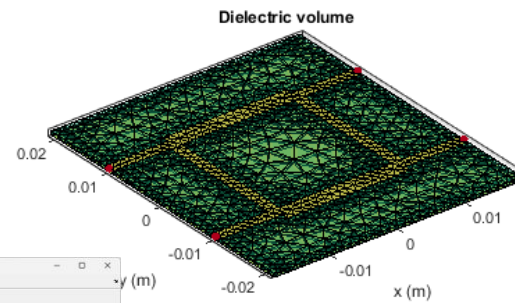
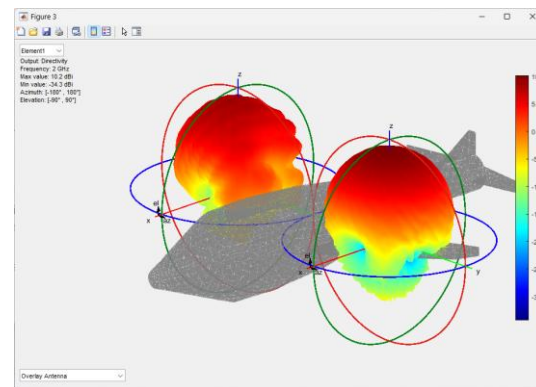
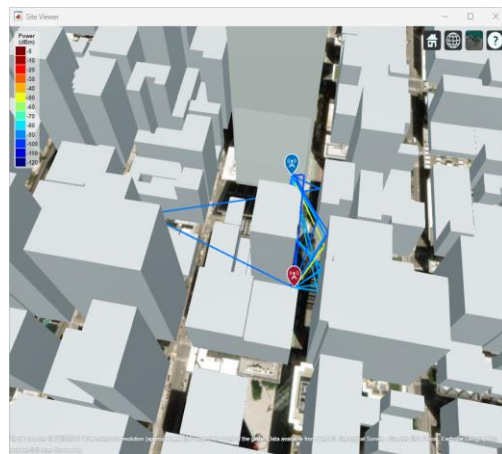
Antenna Workflow #1: Electromagnetic Analysis



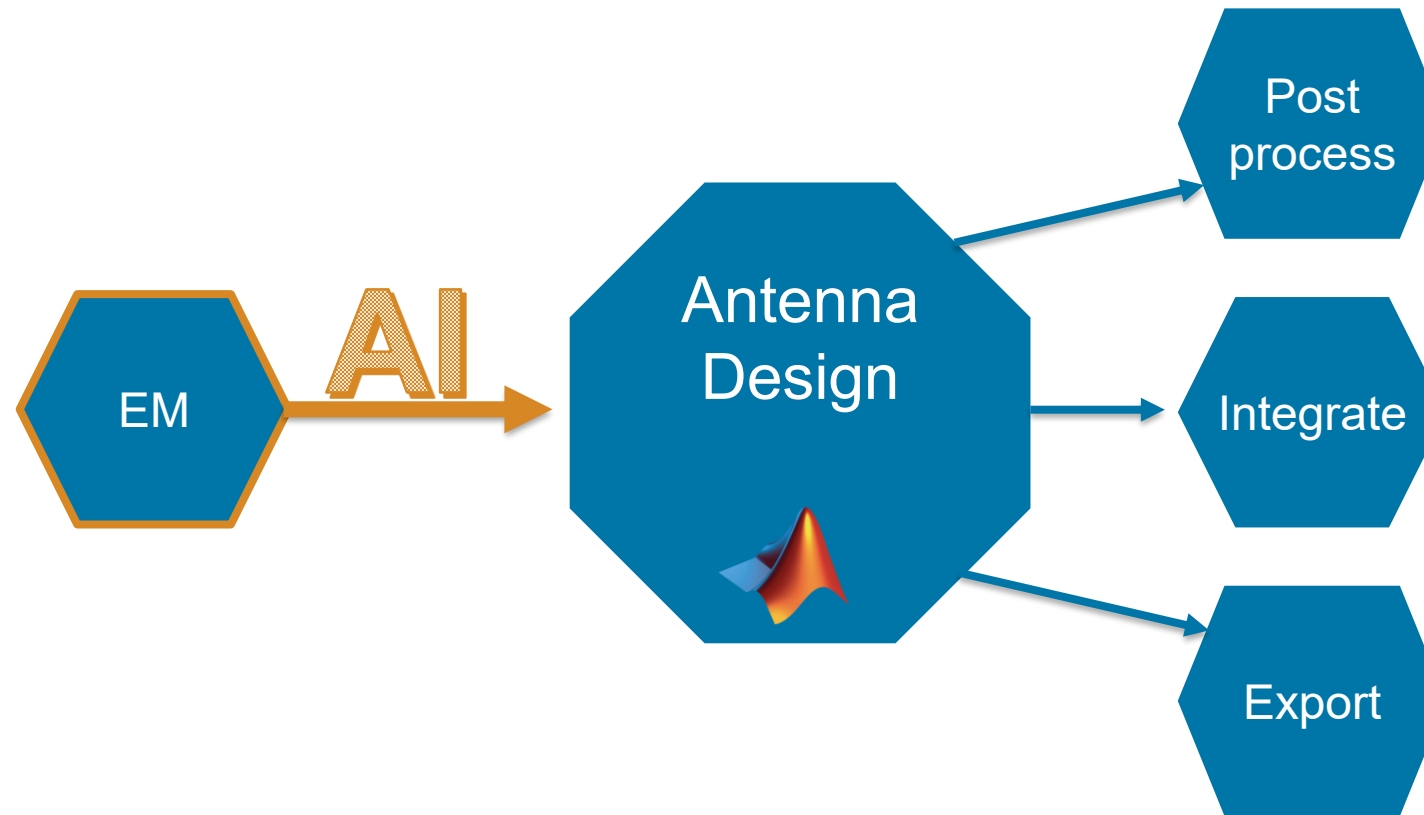
Antenna Toolbox

Design, analyze, and visualize antenna elements and antenna arrays

- Get started with antenna and array catalog and apps
- Perform full-wave electromagnetic analysis (MoM, FMM, ...)
- Design custom planar and 3D structures
- Integrate with RF PCB Toolbox for board design (FEM)
- Analyze the effects of installation on large platforms
- Include RF propagation effects



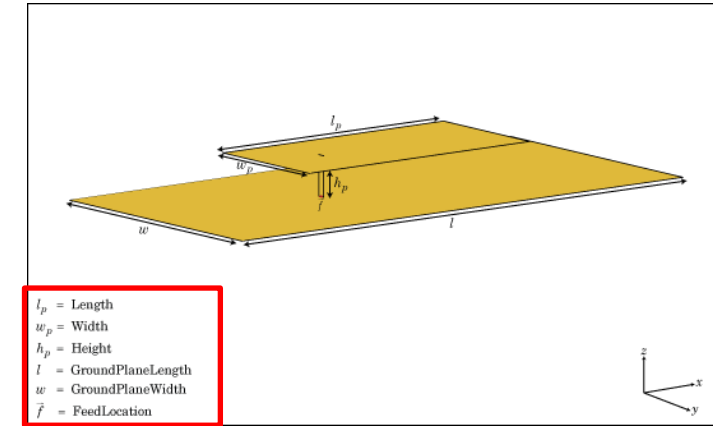
Antenna Workflow #1: Electromagnetic Analysis with AI



- Do you need to explore large design spaces?
- Do you want to speed up antenna analysis with AI?

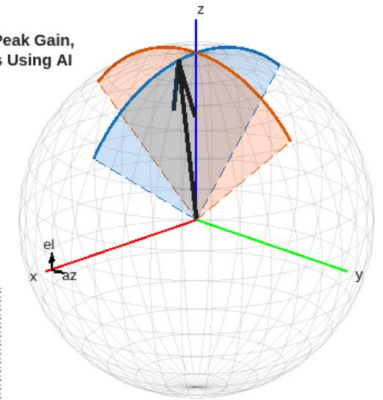
Pretrained AI Antenna Models

- Unique solution within a systematic and extensible framework
- Based on subset of antenna catalog
 - Ready-to-use models, no AI expertise needed
 - Preselected tunable parameters
 - Dipole, dipoleHelix, horn, waveguide, patch antennas (8 types)
 - Substrate materials from dielectric catalog
(supported for 4 antenna types now, and the list is growing)
- Enable AI-based analysis
 - Rapid characterization
 - Design space exploration
 - Port analyses: resonant frequency, bandwidth
 - Field analyses: beamwidth, peak radiation

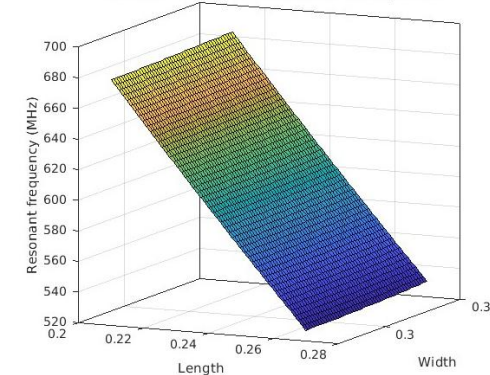


Predict Beamwidth, Peak Gain, and Direction in < 1 s Using AI

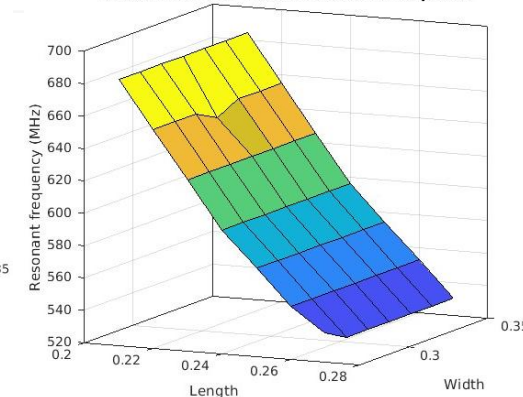
Peak Radiation
 Max Gain: 5.28 dBi
 Azimuth: 3°, Elevation: 81°
Plane 1
 Half-Power Beamwidth: 96°
 Angular Span: [35°, 131°]
Plane 2
 Half-Power Beamwidth: 100°
 Angular Span: [41°, 142°]



Results from AI model -- 16s for 2601 points



Results from ANT model -- 1374s for 49 points



<https://www.mathworks.com/help/antenna/ref/em.ai.aintenna.html>

<https://www.mathworks.com/help/antenna/ug/ai-based-design-space-exp-analysis-and-optimization.html>

S. Sivaramakrishnan, V. Iyer, T. Gao and G. Zucchelli, "A Systematic Application of AI Techniques to Antenna Design, Analysis, and Optimization," 2024 54th European Microwave Conference (EuMC), Paris, France, 2024, pp. 972-975, doi: 10.23919/EuMC61614.2024.10732562

Rapid Design Exploration Using Pretrained AI Models

```
>> p = design(pifa,2.4e9,ForAI=true)
```

```
p =
```

AIAntenna with properties:

Antenna Info

AntennaType: 'pifa'

InitialDesignFrequency: 2.4000e+09

Tunable Parameters

Length: 0.0292

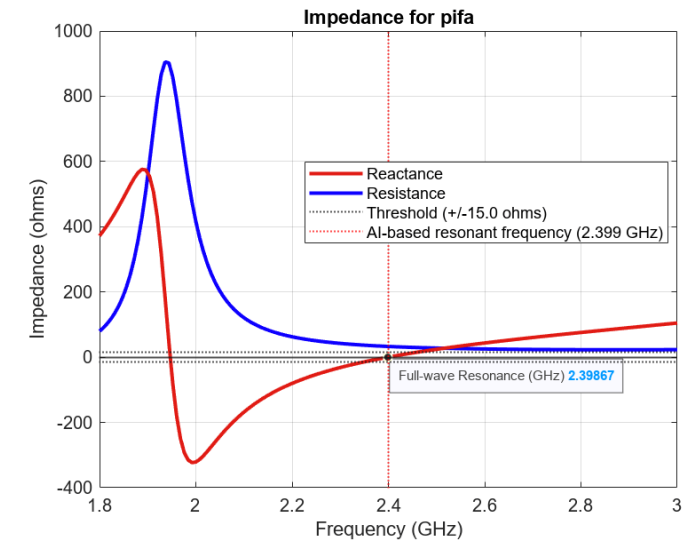
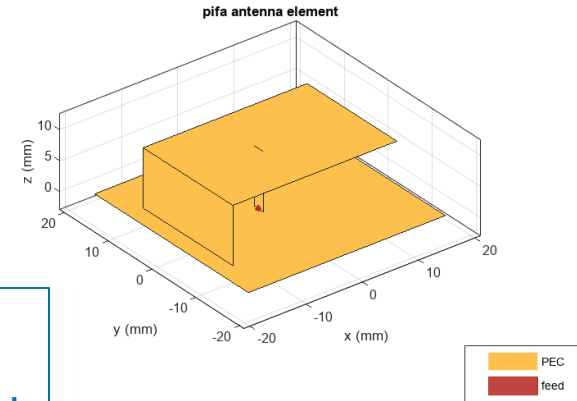
Width: 0.0194

Height: 0.0097

ShortPinWidth: 0.0194

Show read-only properties

Iteratively repeat
this after modifying
tunable parameters!



```
>> tic, fR = resonantFrequency(p), toc
```

```
fR =
```

```
2.4091e+09
```

```
Elapsed time is 0.158043 seconds.
```

```
>> tic, [absBW,fL,fU,matching] = bandwidth(p), toc
```

```
absBW =
```

```
1.4809e+08
```

```
fL =
```

```
2.3186e+09
```

```
fU =
```

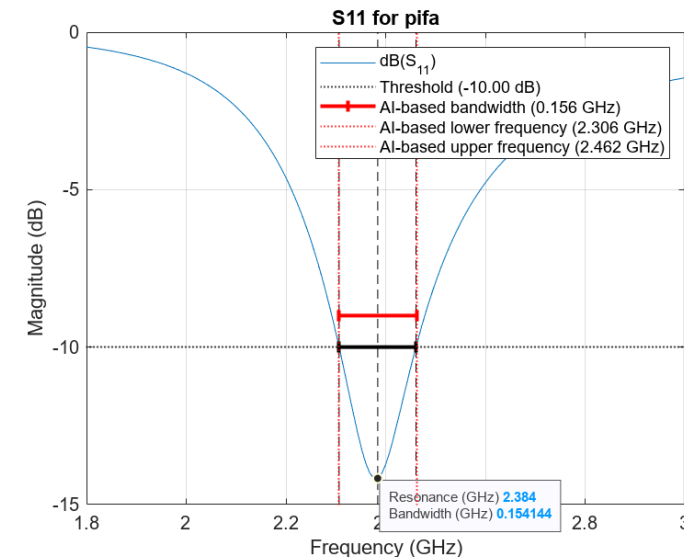
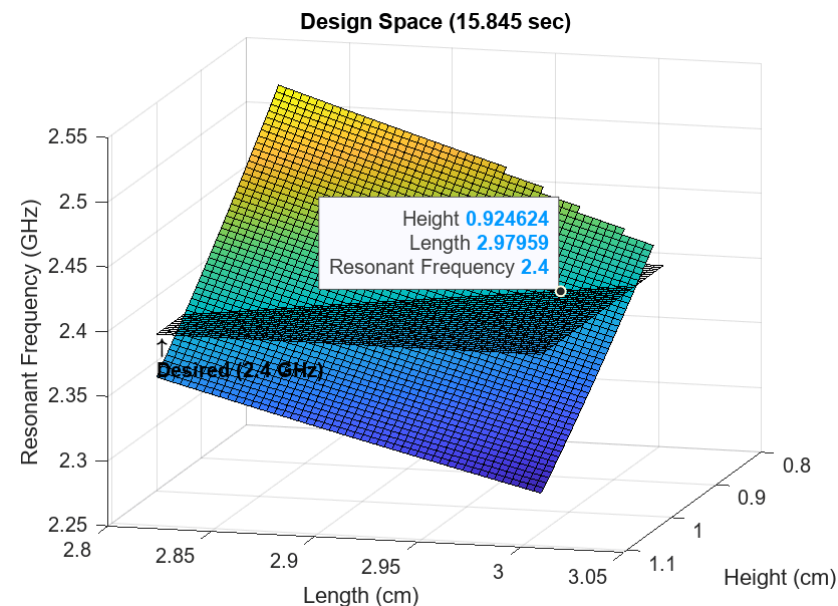
```
2.4667e+09
```

```
matching =
```

categorical

Matched

```
Elapsed time is 0.190911 seconds.
```



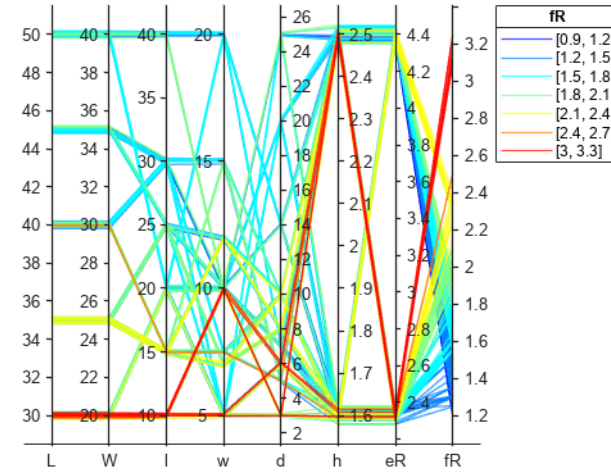
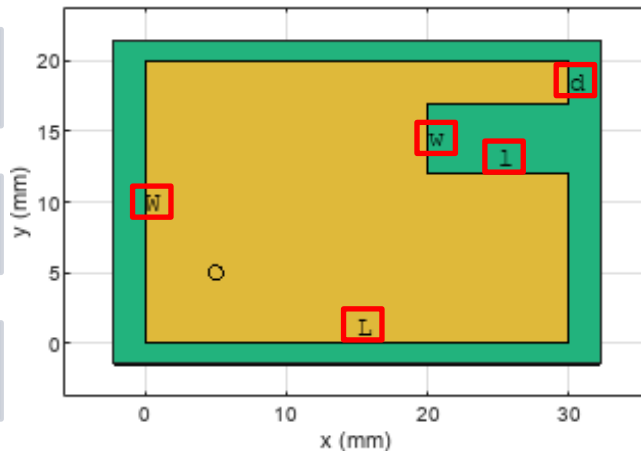
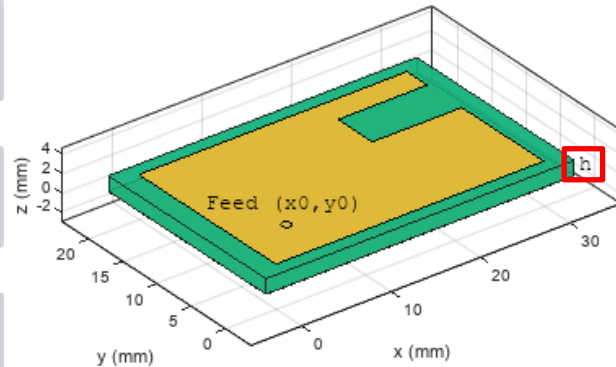
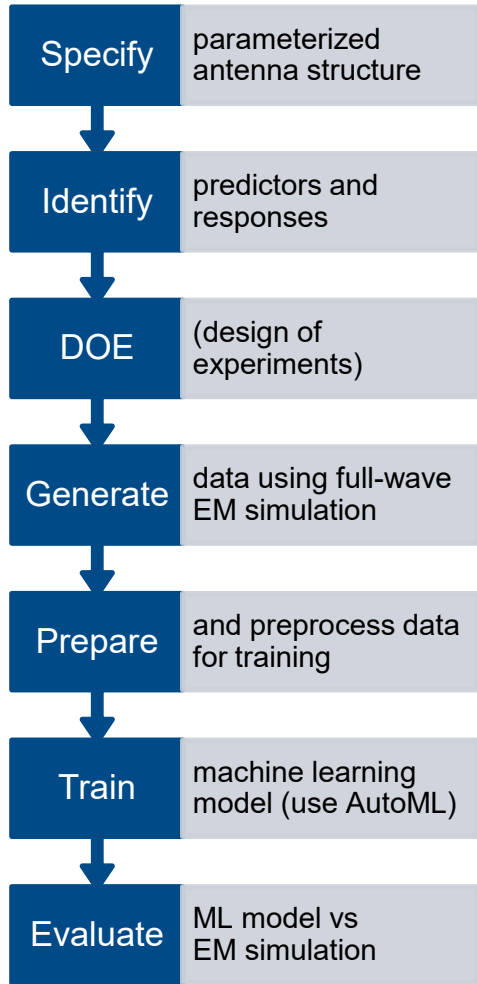
Artificial Intelligence (AI) for Rapid Analysis and Design of Microstrip Patch Antenna

<https://www.mathworks.com/help/antenna/ug/ai-for-rapid-analysis-and-design-of-microstrip-patch.html>

Interactive Exercises Using MATLAB Online R2026a

Live Demo Using R2026b Prerelease

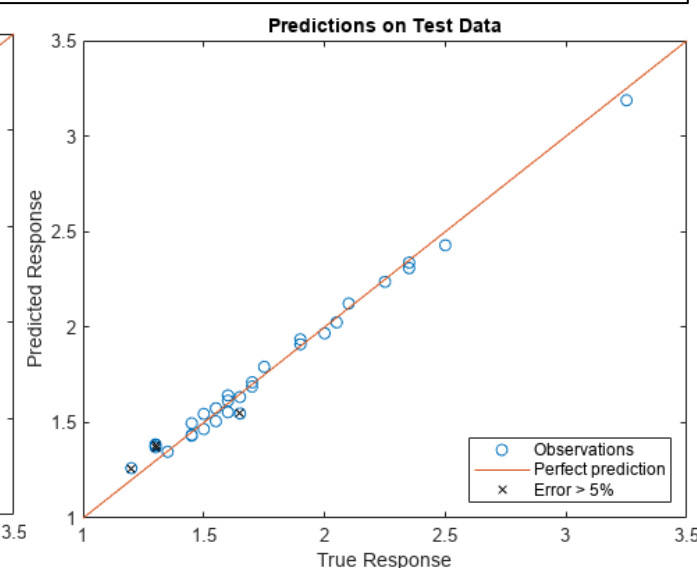
Train your AI Antenna Model



Optimization completed.
 Total iterations: 60
 Total elapsed time: 370.6171 seconds
 Total time for training and validation: 252.4126 seconds

Best observed learner is a gp model with:
 Learner: gp
 Sigma: 0.0073413
 BasisFunction: none
 KernelFunction: ardexpontial
 KernelScale: NaN
 Standardize: false
 Observed log(1 + valloss): 0.023284
 Time for training and validation: 1.2777 seconds

Best estimated learner (returned model) is a gp model with:
 Learner: gp
 Sigma: 0.0073413
 BasisFunction: none
 KernelFunction: ardexpontial
 KernelScale: NaN
 Standardize: false
 Estimated log(1 + valloss): 0.023767
 Estimated time for training and validation: 1.3343 seconds



Train Machine Learning Model for Analysis of Custom Antenna

<https://www.mathworks.com/help/antenna/ug/train-mlmodel-for-analysis-of-custom-antenna.html>